

Surviving funds in illiquid asset management are like those lucky, long-lived smokers. We observe the returns of surviving funds precisely because they are still around, and they are generally above average. All of the unlucky illiquid managers disappear and thus stop reporting returns. Of course, these nonsurvivors have below-average returns.⁹ Industry analysis of buy-out funds, venture capital funds, or [insert your favorite illiquid asset class] tends to encompass only firms that have survived over the period of the analysis. But do we know that the small venture capital firm we're investing in today will be around ten years later? Existing firms and funds, by dint of being alive today, tend to have better-than-average track records. This produces reported returns of these illiquid assets that are too good to be true.

The only way to completely remove the effect of survivorship bias is to observe the entire population of funds. Unfortunately, *in illiquid asset markets we never observe the full universe.*

We can gauge the impact of survivorship bias with mutual funds, which are required to report their returns to the Securities and Exchange Commission because they fall under the 1940 Investment Act. This allows us to see the whole mutual fund universe, at least when the funds become registered, and to compute the effect of survivorship bias. (I provide more details in chapter 16.) Survivorship bias knocks at least 1% to 2% off the estimates of expected returns of mutual funds if we fail to include dead funds in our sample. However, industry often bases its conclusion only on funds in existence at a given point in time. When we separately compare defunct and live funds, the survivorship effect can go above 4%. Take these as lower bounds for illiquid asset managers. Chapters 17 and 18, covering hedge funds and private equity, respectively, show that managers of these investment vehicles often further massage (or manipulate) returns because standardized disclosure is not required and the underlying asset values are not readily observable. In chapter 17, I show that the effect of survivorship and reporting biases for hedge funds is even larger than for mutual funds.

There are data biases other than survivorship bias: for funds specializing in very illiquid assets, reporting returns to database vendors is almost always voluntary. This introduces *reporting biases*.¹⁰ Survivorship bias results when your fund is in the database now and you stop reporting returns because you know your returns are going to be low. Reporting bias also occurs when you don't start reporting your returns in the first place because your fund never achieves a sufficiently attractive track record.

⁹ Jorion and Goetzmann (1999) argue that survivorship bias partly explains the high equity premium (see chapter 8): Countries where we have long histories of equity returns are, by definition, those countries where equity investments have prospered.

¹⁰ See Ang, Rhodes-Kropf and Zhao (2008).